

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.11.06.

1 Sum of Angles in a Quadrilateral

A quadrilateral has three known angles: 85° , 110° , and 95° .

Find the fourth angle. (The angle sum of any quadrilateral is 360° .)

$$360^\circ - 85^\circ - 110^\circ - 95^\circ = 70^\circ. \quad \underline{\hspace{2cm}}$$

2 Rectangle Angles

A rectangle has four angles.

Without measuring, state the measure of each angle and explain why.

Each angle is 90° because a rectangle is defined as having four right angles. Total = $4 \times 90^\circ = 360^\circ$.

3 Parallelogram Angles

A parallelogram has two pairs of equal opposite angles. One angle measures 65° .

Find all four angles.

Opposite angles: 65° and 65° . Adjacent supplementary angles: $180^\circ - 65^\circ = 115^\circ$ each. Angles: 65° , 115° , 65° , 115° . Sum =

4 Find the Unknown

A quadrilateral has angles measuring 3 times the unknown, 2 times the unknown, the unknown itself, and 60° .

Solve: $(3 \times \text{unknown}) + (2 \times \text{unknown}) + \text{unknown} + 60^\circ = 360^\circ$. Find all four angles.

$6 \times \text{unknown} = 300^\circ$; unknown = 50° . Angles: 150° , 100° , 50° , 60° . Sum = 360° .

5 Reasoning: Atlas's Shortcut

Atlas says you can find the missing angle of any quadrilateral by subtracting the three known angles from 360° .

Is this always true? Give an example with angles 82° , 97° , and 104° .

Yes, always true. Missing angle = $360^\circ - 82^\circ - 97^\circ - 104^\circ = 77^\circ$.